

Falling, scoring, dropping

You've completed all the set-up tasks, so it's time to write the code that runs the game. You'll need functions to create the eggs and make them fall, and some more functions to handle egg catches and egg drops.

8 Create the eggs

Add this code. A list keeps track of all the eggs on the screen. The `create_egg()` function decides the coordinates of each new egg (the x coordinate is always randomly selected). Then it creates the egg as an oval and adds it to the list of eggs. Finally, it sets a timer to call the function again after a pause.

```
lives_text = c.create_text(canvas_width - 10, 10, anchor='ne', font=game_font, fill='darkblue', \
                             text='Lives: ' + str(lives_remaining))
```

```
eggs = []
```

This is a list to keep track of the eggs.

```
def create_egg():
```

```
    x = randrange(10, 740)
```

Pick a random position along the top of the canvas for the new egg.

```
    y = 40
```

```
    new_egg = c.create_oval(x, y, x + egg_width, y + egg_height, fill=next(color_cycle), width=0)
```

```
    eggs.append(new_egg)
```

This line of code creates the oval.

```
    root.after(egg_interval, create_egg)
```

The shape is added to the list of eggs.

Call this function again after the number of milliseconds stored in `egg_interval`.

9 Move the eggs

After creating the eggs, add the next function, `move_eggs()`, to set them in motion. It loops through the list of all the eggs on screen. For each egg, the y coordinate is increased, which moves the egg down the screen. Once the egg is moved, the program checks whether it has hit the bottom of the screen. If it has, the egg has been dropped and the `egg_dropped()` function is called. Finally, a timer is set to call the `move_eggs()` function again after a short pause.



```
root.after(egg_interval, create_egg)
```

This line gets each egg's coordinates.

```
def move_eggs():
```

Loop through all the eggs.

```
    for egg in eggs:
```

```
        (egg_x, egg_y, egg_x2, egg_y2) = c.coords(egg)
```

```
        c.move(egg, 0, 10)
```

The egg drops down the screen 10 pixels at a time.

```
        if egg_y2 > canvas_height:
```

Is the egg at the bottom of the screen?

```
            egg_dropped(egg)
```

If so, call the function that deals with dropped eggs.

```
    root.after(egg_speed, move_eggs)
```

Call this function again after the number of milliseconds stored in `egg_speed`.